Supplementary Figures

Supplementary Figure 1

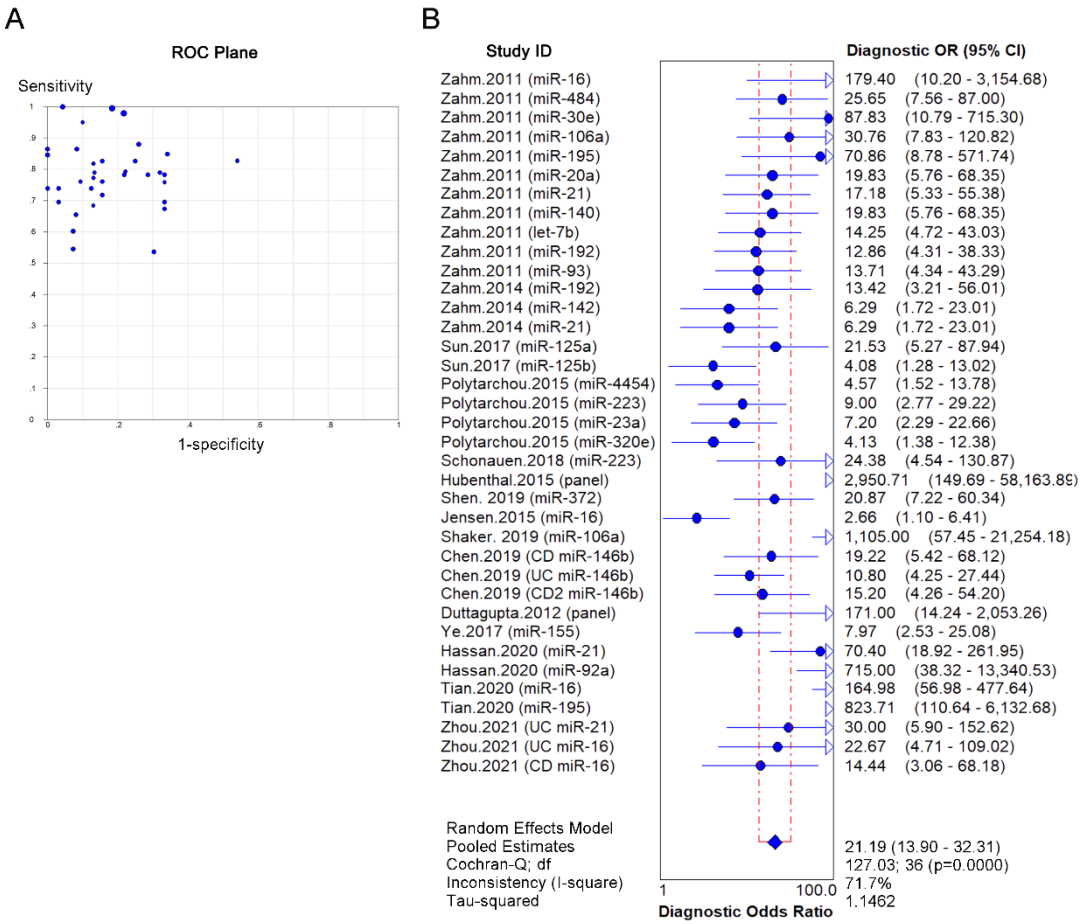

Figure S1 Heterogeneity analysis by the ROC plane and by the diagnostic odds ratio. A, Heterogeneity analysis by the ROC plane. **B,** Diagnostic odds ratio of altered miRNA expression in IBD diagnosis of in all studies. ROC: receiver operating characteristic.

Supplementary Figure 2

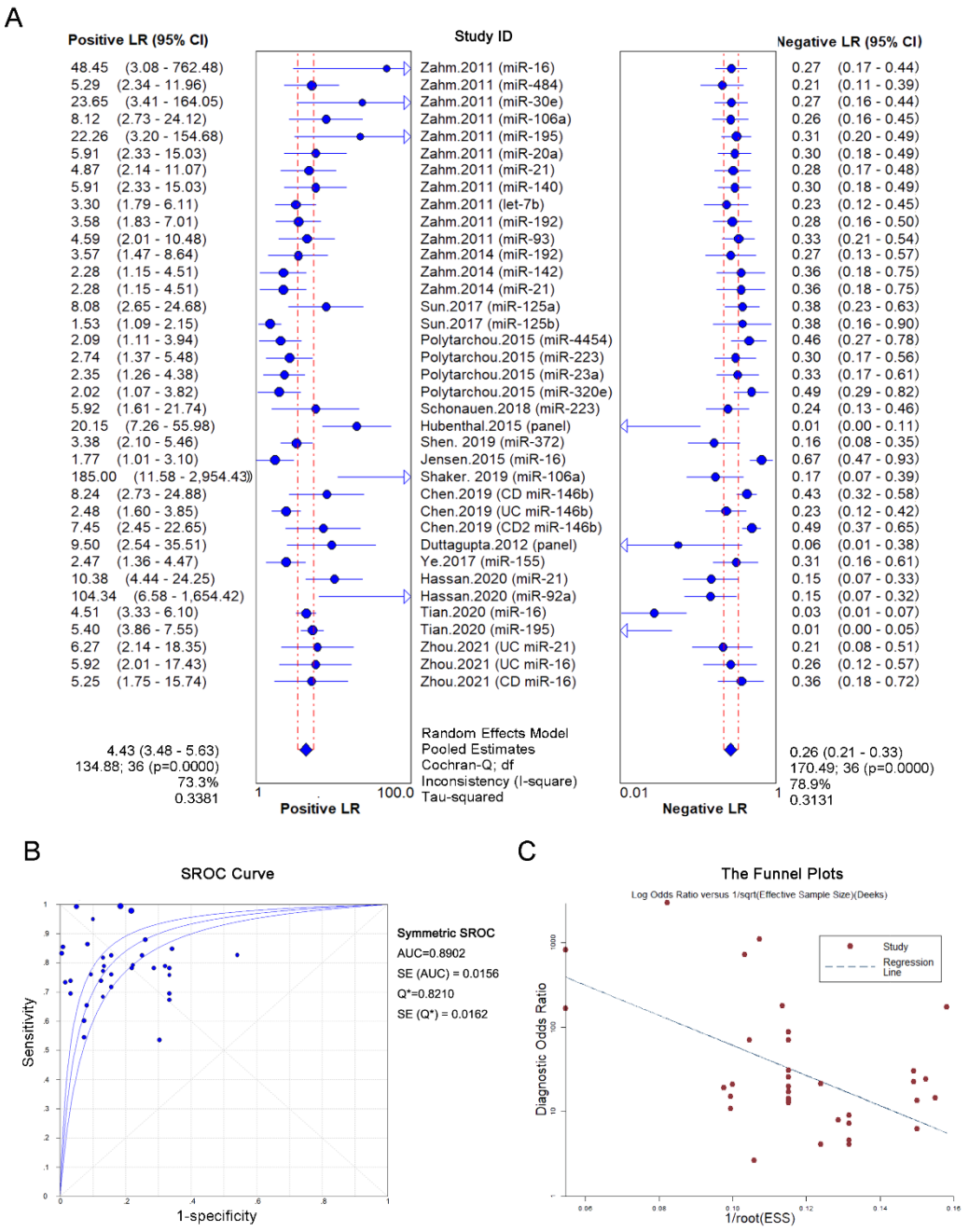

Figure S2 Forest plot of meta-analysis of aberrant expression of microRNAs and its diagnostic value index in IBD, and publication bias assessment by Deeks' funnel plot. A, Positive likelihood ratio and negative likelihood ratio of altered miRNA expression in IBD diagnosis in all studies. **B**, sROC of altered miRNA expression in IBD diagnosis in all studies. **C**, Publication bias assessment by Deeks' funnel plot. IBD: inflammatory bowel diseases; miRNA: microRNAs; sROC: summary receiver operating characteristic.

Supplementary Tables

Table S1 Text-words terms used in searching literature.

No.	MeSH terms/text words used
#1	"Inflammatory Bowel Diseases"[Mesh]
#2	"Crohn Disease"[Mesh]
#3	"Colitis, Ulcerative"[Mesh]
#4	"Enterocolitis"[Mesh]
#5	"Proctitis"[Mesh]
#6	"Ileitis"[Mesh]
#7	#1 OR #2 OR #3 OR #4 OR #5 OR #6
#8	"inflammatory bowel disease*"[All Fields]
#9	crohn*[All Fields]
#10	"ulcerative colitis"[All Fields]
#11	#8 OR #9 OR #10
#12	#7 OR #11
#13	"MicroRNAs"[Mesh]
#14	miR*[All Fields]
#15	miRNA*[All Fields]
#16	MicroRNA*[All Fields]
#17	#13 OR #14 OR #15 OR #16
#18	#12 AND #17

Table S2 Systematic review of included studies that detected altered miRNA expression in patients with IBD using miRNA microarray analysis.

Study ID	Platfrom of miRNA microarray	miRNA probes	Expression validation	Sample from disease/ control (N or pairs)	Cut off	Patients with IBD				Controls				Sample type	The most frequently altered miRNAs (⩾4-fold)		Related clinicopathological factors
						Disease	Gender % (Male/ Female)	Age (y) mean ±SD/ mean(range) /mean	Case number	Control	Gender % (Male/ Female)	Age (y) mean ±SD/ mean(range)/mean	Case number		Over-expressed miRNAs	Down-regulated miRNAs	
Lu Yi Wu 2017	miRCURY LNA microRNA Array	36,333	qPCR	7 CD with inflammation, 7 CD without inflammation and 7 healthy controls	2 folds	Active CD	NA	NA	14	Healthy controls	NA	NA	7	Mucosa	miR-451, miR-31, miR-144, miR-142-5p, miR-142-3p, miRPlus-A1087, miR-374a, miR-338-3p, miR-223, miR-21, miR-126, let-7a	miRPlus-E1186, miRPlus-E1028, miR-1937	NA
Zhen Guo 2015	miRCURY LNA microRNA Array	3100	qPCR	6 pairs CD and 6 healthy controls	2 folds	Active CD	NA	NA	6	Healthy controls	NA	NA	6	Mucosa	miR-4644, miR-4709-3p, miR-5187-3p, miR-4445-5p	let-7g-5p, miR-101-3p, miR-141-3p, miR-145-5p, miR-192-5p, miR-200c-3p, miR-21-5p, miR-22-3p, miR-23b-3p, miR-24-3p, miR-29a-3p, miR-4286, miR-4454, miR-5701, miR-767-5p, let-7b-5p, miR-142-3p, miR-191-5p, miR-200b-3p, miR-	NA

																30c-5p, miR-31-5p, miR-4284	
Aylia Mohammadi 2018	Nanostring technology platform	NA	NA	35 CD, 46 UC and 39 healthy controls	NA	IBD	49.38/50.62	CD: 33±13, UC: 36±10	81	Healthy controls	41.03/58.97	56±9	39	PBMC	Non	Non	IBD type
Yahong Ji 2018	miRCURY LNA microRNA Array	345	qPCR	51 CD, 66 UC and 66 healthy controls	4 folds	IBD	NA	45.19	117	Healthy controls	NA	49.5	66	Feces	miR-199a-5p, miR-223-3p	miR-1226, miR-548ab, miR-515-5p	IBD type, disease severity
Angelos Oikonomopoulos 2016	Nanostring technology platform	NA	NA	45 CD and 21 healthy controls	2 folds	CD	51/49	42±14	45	Healthy controls	38/62	33±8	21	Serum	miR-30e-5p	miR-1183, miR-1286, miR-504	Disease severity, disease location
Feng Wu 2010	Ncode Multi-Species miRNA Microarrays	467	qPCR	11 CD and 13 healthy controls	2 folds	Active CD	72.73/27.27	36.8 (23-64)	11	Healthy controls	46.15/53.85	54.6 (38-68)	13	Mucosa	miR-21, miR-223	Non	IBD type, location
Magali Fasseu 2010	qPCR	321	NA	16 CD and 10 healthy controls	5 folds	IBD	56.25/43.75	41.75 (20-58)	16	Healthy controls	NA	NA	10	Mucosa	miR-126*, miR-29a, miR-29b, miR-324-3p, miR-127-3p, miR-196a, miR-26a, miR-34c-5p, miR-133b, miR-30b, miR-155, miR-106a, miR-22, miR-29c, miR-31, miR-150, miR-21,	miR-320a, miR-215, miR-346, miR-188-5p	IBD type, disease severity

															miR-146b-5p, miR-146a		
M. Iborra 2013	TaqMan® human miRNA array	700	qPCR	18 CD, 18 UC and 33 healthy controls	2 folds	IBD	58.33/41.67	43.08	36	Healthy controls	54.55/45.45	47.3± 9.6	33	Serum and Mucosa	miR-27a*, miR-760, miR-423-5p, miR-128, miR-196b, miR-103, miR-221, miR-532-5p, miR-15b, miR-27a	Non	IBD type, disease severity
Tomohisa Takagi 2010	Ncode human miRNA microarrays	710	qPCR	2 active UC and 2 healthy controls	P < 0.05	Active UC	50/50	35±11	2	Healthy controls	50/50	48±11	2	Mucosa	let-7a, let-7c, let-7d, let-7g, miR-21, miR-155, miR-923	Non	NA
Jan Van der Goten 2014	Affymetrix GeneChip microRNA arrays	4560	qPCR	10 active UC, 7 inactive UC and 10 healthy controls	Top 20	UC	58.82/41.18	Active UC: 55.0 (45.3-61.3), inactive UC: 51.1 (36.1-60.4)	17	Edge biopsies of the ulcers from CDc patients and infectious or ischemic colitis patients	30/70	Active UC: 45.9 (36.1-60.1), inactive UC: 69.1 (68.2-78.4)	10	Mucosa	miR-1971, miR-665, miR-27a-5p, miR-885-3p, miR-3162-5p, miR-31-5p, miR-650, miR-140-5p, miR-223-3p, miR-708-5p, miR-3201, miR-133a, miR-1973, miR-429, miR-196b-3p, miR-147b	miR-422a, miR-141-3p, miR-192-3p, miR-148a-3p	Disease severity
Amy Lewis 2015	miScript miRNA PCR 384HC array	NA	qPCR	6 stricturing CD, 11 nonstricturing CD and 5 healthy controls	NA	CD	NA	NA	17	Healthy controls	NA	NA	5	Serum	Non	Non	NA

Min Min 2014	Human v2.0 miRNA Expression BeadChip	1146	qPCR	20 active UC and 16 healthy controls	1 fold	Active UC	75/25	42.6±12.4	20	Healthy controls	62.5/37.5	52.2±10.0	16	Mucosa	miR-658	Non	NA
Swati Valmiki 2017	GeneChip® miRNA 4.0 Array	NA	qPCR	8 UC and 8 non-IBD controls	1.5 folds	UC	37.5/62.5	39.75±10.29	8	NIBD-related illnesses	25/75	40.75±11.86	8	Mucosa	miR-138-5p, miR-708-5p, miR-212-3p, miR-4538, miR-4521, miR-4417, miR-17-3p, miR-424-3p, miR-874-3p, miR-25-5p, miR-223-3p, miR-1271-5p, miR-148b-3p, miR-501-5p, miR-4486, miR-224-3p, miR-21-3p, miR-146b-5p, miR-149-5p, miR-31-5p	miR-552-3p, miR-196b-5p, miR-378d-5p	NA
Jeremy S Schaefer 2015	miRCURY LNA microRNA Array	over 600	qPCR	2 CD, 2 UC and 1 healthy control	3 folds	IBD	NA	NA	4	Healthy controls	NA	NA	1	Mucosa	miR-31, miR-494	Non	NA
Bailey C. E. Peck 2015	Illumina HiSeq 2500 platform	NA	qPCR	21 CD, 6 UC and 14 healthy controls	2 folds	Inactive CD	33/67	36.6 (15-76)	21	NIBD-related illnesses	57/43	56.4 (41-82)	14	FFPE	NA	NA	Disease subclass (about stricturing and penetrating)
Mehmet Cosk	NA	NA	qPCR	4 CD, 4 UC and 2	NA	IBD	50/50	37 (25-73)	8	Healthy controls	50/50	39 (37-41)	2	Mucosa	miR-15a, miR-199b-3p, miR-20b, miR-20a, miR-106b, miR-	Non	IBD type, diseas

un 2013				healthy controls											27b, miR-99a, miR-222, miR-151-5p, miR-203, miR-30a, miR-25, miR-26b, miR-646, miR-125b, miR-98, miR-768-3p, miR-195, miR-99b, miR-23a, miR-18a, miR-17, miR-155, miR-23b, miR-1201, miR-130a, miR-199a-3p, miR-16, miR-126, miR-106a, miR-1248, miR-27a, miR-222*, miR-506, miR-125b-1*, let-7e*, miR-512-5p, miR-1288, miR-330-3p, miR-623, miR-34b		e severity
María Rojas-Feria 2018	NA	88	qPCR	NA	1.5 folds	Active CD	NA	NA	17	Paired adjacent normal colon tissues	NA	NA	18	Mucosa	miR-144, miR-519	Non	Gut microbiota
Zhen Guo 2016	miRCURY LNA microRNA Array	NA	qPCR	6 pairs CD and 6 healthy controls	2 folds	Active CD	NA	NA	6	Healthy controls	NA	NA	6	Mucosa	Non	miR-192-5p	NA
Jingmei Lin 2016	Illumina next generation sequencing of small RNA	NA	qPCR	9 CD, 10 UC and 18 healthy controls	2 folds	IBD	NA	NA	19	Healthy controls	NA	NA	18	Mucosa	miR-194-2, miR-383, miR-615	Non	Disease severity

Geoffrey W. Kristiansen 2015	NA	289	qPCR	57 CD, 62 UC and 58 healthy controls	2 folds	IBD	50.42/49.58	40 (16-74)	119	Healthy controls	44.83/55.17	43.7 (19-67)	58	Serum	miR-595, miR-143, miR-1246	Non	IBD type, disease severity
Feng Wu 2011	miRCURY LNA microRNA Array	1300	qPCR	19 CD, 23 UC and 13 healthy controls	miRNA signal ＞ mean±2 SD of the controls	IBD	47.62/52.38	46.5 (21-81)	42	Healthy controls	46.2/53.8	56.3 (43-75)	13	Blood	miR-362-3p	miR-505*	IBD type, disease severity
Raju Ranjha 2015	miRCURY LNA microRNA Array	NA	qPCR	30 UC and 20 non-IBD controls	1.7 folds	UC	56.7/43.3	34.84±13.75	30	NIBD-related illnesses	65/35	34.23±13.1	20	Mucosa	miR-141-3p	Non	Location
Georgios Koukos 2015	miRCURY LNA microRNA Array	NA	qPCR	5 pairs UC	NA	UC	40/60	15.2±1.83	5	Paired adjacent normal colon tissues	40/60	15.2±1.83	5	Mucosa	miR-223, miR-1973, miR-3611, miR-21	miR-4284, miR-378a/c/d, miR-378a, miR-138-1, miR-4286	Age
Jing Han 2018	miRCURY LNA microRNA Array	NA	qPCR	3 UC and 3 healthy controls	NA	Active UC	NA	NA	3	Healthy controls	NA	NA	3	Mucosa	Non	Non	NA
Karen Dubois-Cama	Affymetrix GeneChip microRNA arrays	more than 5,000 mature miRNAs	qPCR	8 pairs UC	1.5 folds	Active UC	37.5/62.5	34 (18-44)	8	Paired adjacent normal colon tissues	37.5/62.5	34 (18-44)	8	Mucosa	miR-223-3p, miR-708-5p	miR-147b, hsa-miR-196b-3p	Disease severity

cho 2019																	
Aylia Mohammadi 2019	Nanostring technology platform	798	Non	23 quiescent CD and 38 healthy controls	P＜0.05	Inactive CD	34.78/64.22	34±13	23	Healthy controls	57.89/42.11	55±9	38	Mucosa	Non	Non	Location
Julien Verdier 2020	nCounter human v2 miRNA Expression Assay	800	qPCR	6 CD and 6 healthy controls	P＜0.05	Active CD	83/17	37.6 (27-69)	6	Healthy controls	83/17	36.8 (25-73)	6	Feces	miR-223, miR-1246, miR-451, miR-21, miR-15a, miR-15b	Non	Disease severity
Filip Ambrozkiewicz 2020	Nanostring technology platform	800	Non	15 CD and 9 healthy controls	1.5 folds	CD	66.7/33.3	32 (20-62)	15	Healthy controls	NA	NA	9	Feces	miR-223-3p	miR-577, miR-642a-3p, miR-26b-5p	Disease severity
Christian T. Wohnhaas 2020	Nanostring technology platform	555	qPCR	52 CD and 15 healthy controls	3.0 folds	CD	25/75	38 (19–74)	52	Healthy controls	53.3/46.7	38 (23–65)	15	Feces	miR-16-5p, miR-142-5p, miR-223-3p	miR-10b-5p, miR-192-5p, miR-10a-5p, miR-375	Disease severity
Jie Zhao 2020	Agilent Human miRNA Microarray	NA	qPCR	3 CD and 3 healthy controls	4.0 folds	CD	NA	54.3±4.8	3	Healthy controls	NA	54.1±5.1	3	Mucosa	miR-130a-3p, miR-106b-5p, miR-30b-5p	Non	NA
Kallar 2021	miRCURY LNA microRNA Array	340	qPCR	9 CD, 14 UC, 1 IBDU and 8 healthy controls	NA	IBD	66.67/33.33	34 (18-68)	24	Healthy controls	50/50	43 (20-59)	8	Leucocytes (CD4+, CD8+ and CD14+ cells)	CD4+ cells: miR-4792, miR-921, CD8+ cells: miR-4792, CD14+ cells: non	CD4+ cells: non, CD8+ cells: miR-200b-3p, CD14+ cells: non	NA

miRNA: microRNAs; IBD: Inflammatory bowel disease; UC: Ulcerative colitis; CD: Crohn's disease; y: years; SD: Standard deviation; PBMC: Peripheral blood mononuclear cell; FFPE: Formalin-fixed paraffin-embedded; qPCR: Quantitative real-time polymerase chain reaction; NA: Not available; Non: None of valid data.

Table S3 Systematic review of included studies that detected altered miRNA expression in patients with IBD using validated methods.

Study ID	Patients with IBD				Controls				Sample type	miRNA assay	Normalization controls	No. of miRNAs	Up or down	miRNA studied	Target	Related clinicopathological factors
	Disease	Gender % (Male/ Female)	Age (y) mean±SD/ mean(range)/mean	Case number	Control	Gender % (Male/ Female)	Age (y) mean±SD/ mean(range)/mean	Case number								
Benjamin P. Keith 2018	CD	50.22/49.78	26.85±20.27	229	Non-IBD	NA	NA	187	Mucosa	qPCR	RNU48	1	Up	miR-31	NA	Disease location, ileal stricturing
Georgios Koukos 2013	IBD	NA	Adult and children	33	Non-IBD	NA	NA	12	Mucosa	qPCR	U6	1	Down	miR-124	STAT 3	Age, IBD type, disease severity
Hye-Youn Kim 2016	Active UC and CD	NA	NA	10	Healthy controls	NA	NA	5	Mucosa	qPCR	U6	2	Up	miR-132, miR-223	FOXO 3a	IBD type
Mao Cai 2017	Active UC	66.67/33.33	50.4±13.8	15	Healthy controls	69.23/30.78	43.1±11.2	13	Mucosa	qPCR	U6	1	Down	miR-141	CXCL 5	NA
Jing Han 2018	Active UC	NA	NA	NA	Healthy controls	NA	NA	NA	Mucosa	qPCR	NA	1	Up	miR-142-5p	SOCS 1	NA
Zhen Huang 2014	Active CD	NA	NA	15	Paired adjacent normal colon tissues	NA	NA	15	Mucosa	qPCR	U6	1	Down	miR-141	CXCL 12β	Histological score
Guodong Chen 2015	Active UC	42.86/57.14	42.5	35	Healthy controls	50/50	34±8.9	10	Mucosa and serum	qPCR	U6	1	Down	miR-195	Smad 7	Steroid resistance
Nina Zidar 2016	Active UC and CD	UC: 40/60, CD: 40/60	UC: 50.6±18.28, CD: 41.7±13.1	UC: 10, CD: 10	Paired macroscopically normal mucosa	40/60	UC: 50.6±18.28, CD: 41.7±13.1	UC: 10, CD: 10	Surgery tissue	qPCR	U6	5	Down	miR-141, miR-200a, miR-200b, miR-200c and miR-429	NA	IBD type
Ye Zhao 2016	Active CD	65/35	32.05±10.59	20	Healthy controls	75/25	35.1±15.68	20	Mucosa	qPCR	U6	4	Up or down	Up: miR-124, miR-9, down: miR-145, miR-143	AHR (miR-124)	NA

Huixia Zhang 2018	Active UC	NA	NA	23	Healthy controls, IBS	NA	NA	20	Mucosa	qPCR	U6	1	Up	miR-15	A2aAR	NA
Chen Zhang 2014	Active UC and CD	UC: 72.2/27.8, CD: 61.5/38.5	UC: 46 (19-74), CD: 41 (16-68)	UC: 36, CD: 26	Diverticular disease	73.7/26.3	50 (23-87)	38	FFPE	qPCR	U6	1	Up	miR-31	NA	IBD type
Bin Zhang 2017	IBD	NA	NA	NA	Paired adjacent normal colon tissues	NA	NA	NA	Surgery tissue	qPCR	NA	1	Up	miR-122a	EGFR	NA
Yongzhi Yang 2013	UC	46.67/53.33	39.4 (20-65)	15	Healthy controls	53.33/46.67	44.5 (23-67)	15	Mucosa	qPCR, ISH	U6	1	Up	miR-21	RhoB	NA
Xue Yang 2018	Active UC and CD	NA	NA	UC: 32, CD: 35	①Healthy controls, ②inactive IBD and ③paired adjacent normal colon tissues	NA	NA	①43, ② UC: 28, CD: 29 and ③ UC: 32, CD: 35	Mucosa, PBMC	qPCR	U6	1	Up	miR-425	Foxo1	Disease severity
Weiyun Wu 2017	Active UC	NA	NA	26	Healthy controls	NA	NA	19	Mucosa	qPCR	U6	1	Up	miR-206	A3AR	NA
Simon R Whiteoak 2018 (Cohort 1)	Active UC	52/48	49 (36-62)	25	①Healthy controls，②inactive UC patients	①47.62/52.38，②53.33/46.67	①59 (49-66), ②55 (49-66)	①21, ②19	Mucosa	qPCR	RNU44	1	Up	miR-31	TSLP	Disease severity
Simon R Whiteoak 2018 (Cohort 2)	Active UC	25/75	53 (46-62)	4	Healthy controls	25/75	76 (61-89)	4	Mucosal lymphocytes	qPCR	RNU44	1	Up	miR-31	TSLP	NA
Simon R Whiteoak 2018 (Cohort 3)	Active UC	80/20	50 (31-66)	5	Healthy controls	66.67/33.33	64 (35-71)	6	Blood ($CD4^+CD25^{intermediate}$ T cells)	qPCR	RNU44	1	Up	miR-31	TSLP	NA

Gorm Thorlacius-Ussing 2017	Active UC and CD	UC: 40/60, CD: 87.5/12.5	UC: 46.1 (32-67), CD: 54 (29-77)	UC: 10, CD: 8	Healthy controls	55.56/44.44	54.2 (26-83)	9	FFPE	①qPCR, ②Quantitative ISH	①U6, hsa-let-7a-5p, hsa-miR-103a-3p, and hsa-miR-191-5p, ②scramble probe	①3，②2	Up	①miR-21, miR-126, miR-223, ②miR-21, miR-126	NA	IBD type
Wen-Juan Tang 2018	Active UC and CD	UC: 57.14/42.86, CD: 52.63/47.37	UC: 9.74±2.96, CD:10.76±3.25	UC: 7, CD: 19	①Juvenile polyps, ②Inactive CD patients	①57.14/42.86②57.14/42.86	①9.57±3.51, ②11.86±3.25	①21, ②7	Mucosa	qPCR	U6	1	Up	miR-15a	Cdc42	IBD type, disease severity
Dániel Szűcs 2016	Active CD	80/20	12.11±1.63	10	①Non-IBD, ②inactive CD patients	①80/20, ②50/50	① 8.75±2.36, ② 12.4±1.52	①10, ②10	Mucosa	qPCR	U6	3	Up	miR-146a, miR-155, miR-122	NA	Disease severity
Magali Svrcek 2013	IBD	50/50	37 (19-78)	18 (24 samples)	Healthy mucosa from patients with diverticulitis	30/70	58.6 (24-81)	20	Mucosa	qPCR	RNU48	3	Up	miR-155, miR-21, miR-423-5p	NA	CRC
Xu-Feng Pei 2018	IBD	UC: 46.30/53.70, CD: 53.33/46.67	UC: 36.51±8.14, CD: 33.58 ± 7.36	UC: 54, CD: 45	Healthy mucosa from patients with diverticulitis; peripheral venous blood was taken from healthy examinees	53.33/46.67; NA	35.14± 6.45, NA	Mucosa: 15, peripheral venous blood: 20	Mucosa and blood ($CD4^+$ T cells)	qPCR	U6	1	Up	miR-22	HDAC4	IBD type, disease severity
Surajit Pathak 2015	Active UC and CD	UC: 37.5/62.	UC:42 (24-71),	UC: 8, CD:8	Healthy controls	50/50	55 (27-69)	8	Mucosa (intestinal	qPCR	U6	1	Up	miR-155	SOCS1	NA

		5, CD: 50/50	CD:39 (28-64)					fibroblasts and myofibroblasts)							
Archanioti Paraskevi 2012	Active UC and CD	UC: 47.73/52.27, CD: 44.53/55.47	UC: 33.36±11.23, CD: 30.72±10.61	UC: 88, CD: 128	Healthy controls	46.92/53.08	44.52±8.64	162	Blood	qPCR	U6	19	Up	UC: miR-16, miR-21, miR-28-5p, miR-143, miR-151-5p, miR-155, miR-199a-5p, CD: miR-16, miR-23a, miR-28-5p, miR-29a, miR-106a, miR-107, miR-126, miR-191,miR-199a-5p, miR-200c, miR-362-3p, miR-532-3p	NA	IBD type
Ameneh Omidbakhsh 2018	IBD	UC: 53.12/46.88, CD: 53.12/46.88	UC: 31.53±12.28, CD: 32.45±11.3	UC: 32, CD: 32	Healthy controls	53.12/46.88	33.47±10.92	32	Blood	qPCR	U6	2	Up	miR-106a, miR-362-3p	NA	IBD type, disease severity
Alexandru V. Olaru 2011	IBD	68.57/34.43	46.2	35	Normal mucosa from patients without IBD or colorectal cancer history	85.71/14.29	60.6	14	Mucosa	qPCR	U6	1	Up	miR-31	FIH-1	Disease severity, neoplasia, CRC
Viola Neudecker 2017	Active UC and CD	NA	UC:51.5, CD: 36.7	UC: 6, CD: 6	①Healthy controls, ②inactive	NA	①51.4, ② UC: 47.8, CD: 47.5	①10, ② UC: 6, CD: 4	Mucosa	qPCR	U6	1	Up	miR-223	NA	IBD type, disease severity

					patients with IBD											
Hang Thi Thu Nguyen 2014	IBD	NA	NA	UC: 20, CD: 20	Non-IBD	NA	NA	13	Mucosa	qPCR	NA	2	Up	miR-130c, miR-130a	NA	IBD type, disease severity
Mousa Mohammadnia-Afrouzi 2016	UC	46.67/53.33	37.26±9.59	30	Healthy controls	50/50	36.90±10.61	30	Blood ($CD4^+CD25^+CD127^{-/low}FoxP3^+$ Treg cells)	qPCR	RNU48	8	Up or down	Up: miR-29b, miR-31, miR-181a, down: miR-21, miR-29a, miR-146a, miR-155, miR-181c	NA	NA
Carlos D. Minacapelli 2019	Active UC	53.33/46.67	41.73 (23-64)	15	Healthy controls	60/40	40.8 (42-65)	5	Mucosa	qPCR	β-actin	1	Up	miR-206	NA	Efficacy of 5-aminosalicylic acid
Friederike Cordes 2016	CD	NA	38 (20-55)	15	Healthy controls	NA	30 (18-65)	13	Blood	qPCR	U6	1	Up	miR-320a	PPP2R5B, JAM-A	Disease severity
Weixu Chen 2014	Active CD	NA	NA	10	Healthy controls	NA	NA	10	Mucosa	qPCR	U6	1	Up	miR-124	AHR	NA
Shameer J. Mehta 2018	CD	50/50	36.02	42	Healthy controls	NA	NA	13	Serum	qPCR	U6	4	Down	miR‐141, miR‐200a, miR‐200b and miR‐200c	ZEB1	Stricturing
Nitsan Maharshak 2013	IBD	34/66	13.45±39.9	33	Healthy controls	44/56	7.93±51.2	16	Mucosa	qPCR	SNO135	1	Up	miR-132	AChE	Disease severity
Dan Ma 2019	Active UC and CD	NA	NA	16	Paired adjacent normal colon tissues	NA	NA	11	Mucosa	qPCR	U6	1	Down	miR-185-3p	MLCK	NA
Jin-an Li 2017	UC	NA	NA	24	Healthy controls	NA	NA	20	Mucosa	qPCR	U6	1	Down	miR-214-3p	STAT6	NA

Ying Kang 2016	IBD	UC: 57.14/42.86, CD: 54.05/45.95	UC: 41.6±10.4, CD: 38.3±9.2	UC: 63, CD: 37	Healthy controls	59.52/40.48	40.5±12.4	42	Mucosa and serum	qPCR	U6	1	Up	miR-595	NCAM1, FGFR2	IBD type, disease severity
Chong He 2016	IBD	NA	NA	UC: 66, CD: 72	Healthy controls	NA	NA	32	Mucosa	qPCR	U6	1	Up	miR-301a	SNIP1	Disease severity
Markus Gwiggner 2018	Active UC	54.55/45.45	47.3 (22-85)	11	Healthy controls	54.55/45.45	56.1 (46-78)	11	Mucosa	qPCR	RNU44	4	Up or down	Up: miR-31-5p, miR-155-5p, miR-183-5p, down: miR-324-3p	IL13RA	IBD type, disease severity
Jing Guo 2017	IBD	87.50/12.50	3.5-14	UC: 3, CD: 5	Normal colon tissues from patients who had undergone intestinal surgery for other diseases	62.50/37.50	2.75-10	8	Mucosa	qPCR	U6	1	Up	miR-7-5p	TFF3	NA
Sezin Günaltay 2014	UC	87.50/12.50	56.38 (33-79)	16	Non-IBD	54.55/45.45	58.6 (29-88)	11	Mucosa	qPCR	U6, RNU44 and RNU48	3	Up	miR-146a, miR-155, miR-21	NA	Disease severity
Y Chen 2013	Active UC and CD	NA	NA	UC: 11, CD: 11	Paired adjacent normal colon tissues	NA	NA	UC: 11, CD: 11	Mucosa	qPCR	U6	1	Down	miR-200b	ZEB1	IBD type
Xu-tao Lin 2018	Active CD	NA	NA	18	Paired adjacent normal colon tissues	NA	NA	18	Mucosa	qPCR	U6	1	Up	miR-143	ATG2B	NA
Tanzhou Chen 2017	IBD	NA	NA	20	Non-IBD	NA	NA	5	Mucosa	qPCR	NA	1	Up	miR-126	S1PR2	NA

Bin Chen 2013	Active UC	62.5/37.5	36.5 (19-51)	24	Healthy controls	60/40	32.4 (21-48)	10	Mucosa	qPCR	U6	1	Down	miR-19a	TNF-a	NA
Chong He 2017	IBD	NA	NA	UC: 72, CD: 81	Healthy controls	NA	NA	35	Mucosa (intestinal epithelial cells)	qPCR	U6	1	Up	miR-301a	BTG1	IBD type, disease severity
Nóra J. Béres 2016 (Cohort 1)	CD	50/50	14.73±0.58	12	Healthy controls	68.75/31.25	10.56±1.36	16	FFPE	qPCR	U6	3	Up	miR-146, miR-155, miR-122	NA	Disease severity
Nóra J. Béres 2016 (Cohort 2)	Active UC and Active or inactive CD	UC: 60/40, CD: 44.74/55.26	UC: 11.8±1.75, active CD: 12.92±1.13, inactive CD: 12.86±1.6	UC: 10, CD: 38	Healthy controls	69.57/30.43	8.57±1.09	23	Mucosa	qPCR	U6	3	Up	miR-146, miR-155, miR-122	NA	IBD type, disease severity
Nóra Judit Béres 2018	CD	NA	Active: 12.19±1.2, inactive: 10.97±1.29	33	Non-colitis-controls	NA	6.48±1.1	20	Mucosa	qPCR	U6	1	Up	miR-223	NA	Disease severity
Natalya Benderska 2015	UC	40/60	42.5±14.5	10	Healthy controls	46.67/53.33	41.6±31	15	Mucosa	qPCR	U6	1	Up	miR-26b	DIP1	NA
Yugo Ando 2016	IBD	UC: 73.68/26.32, CD: 58.82/41.18	Active UC: 45.6±14.6, inactive UC: 49.7±14.2, active CD: 46.0±19.4, inactive CD: 37.2±15.9	UC: 19, CD: 17	Non-IBD	40/60	54.9±16.1	10	Mucosa ($CD3^+$ T cells)	qPCR	SNORD95	1	Down	miR-21	NA	IBD type, disease severity

Xiao Feng 2012	UC	40.91/59.09	active: 48.8 (17-80), inactive: 37(25-58)	22	Healthy controls, IBS	Healthy controls: 46.7/53.3. IBS: 60/40	Healthy controls: 47.1±54.61, IBS:40.2±38.3	Healthy control: 15, IBS: 15	Mucosa	qPCR	U6	2	Up	miR-126, miR-21	miR-126: IKBA, PLK2, CRK	Disease severity
Gao Chao 2019	IBD	NA	NA	UC: 31, CD: 37	Healthy controls	NA	NA	24	Blood ($CD4^+CD25^+Foxp3^+$ Treg cells)	qPCR	U6	1	Up	miR-155	CTLA-4	IBD type, disease severity
Min Chen 2019	Active CD	70/30	31-51	10	Normal tissue of anastomosis patients	70/30	31-51	10	Surgery tissue	qPCR	NA	1	Up	miR-16-1	HSP70	NA
Richard K. Felwick 2019	CD	47.83/52.17	active: 32(17-66), inactive: 51(26-80)	23	Healthy controls	30/70	70 (49-86)	10	Mucosa (epithelial cells)	qPCR	RNU44	1	Up	miR-23a	TNFAIP3	Disease severity
Yingqi Shi 2019	IBD	UC: 60/40, CD: 52/48	UC: 38.77±9.21, CD: 34.15±7.68	UC: 40, CD: 24	Healthy controls	52.31/47.69	41.62±9.45	65	Mucosa, serum and PBMC	qPCR	NA	1	Down	miR-10a	NA	IBD type
Artin Soroosh 2019	IBD	NA	NA	UC: 57, CD: 22	Healthy controls	NA	NA	26	Mucosa	qPCR	U6	1	Up	miR-24	NA	IBD type, disease severity
CP Wu 2019	IBD	UC: 57.14/42.86, CD: 50/50	UC: 26-66, CD: 23-50	UC: 14, CD12	Healthy controls	NA	NA	30	Mucosa	qPCR	U6	1	Down	miR-375	TLR4	IBD type
Xiaohong Lu 2020	UC	NA	NA	45	Healthy controls	NA	NA	45	Serum	qPCR	U6	1	Up	miR-21-5p	STAT3	NA
Yan Shi 2020 (Cohort1)	IBD	UC: 41.18/58.82, CD:	UC: 47.6±2.92,	UC: 17, CD: 15	Healthy controls	42.86/57.14	29.2±1.01	14	Mucosa	qPCR	sNANA 5s	1	Down	miR-219a-5p	NA	IBD type, disease severity

		66.67/33.33	CD: 37.5±3.23													
Yan Shi 2020 (Cohort2)	IBD	UC: 55.56/44.44, CD: 60/40	UC: 46.8±4.31, CD: 38.8±3.25	UC: 18, CD: 15	Healthy controls	56.52/43.48	29.33±1.13	23	Blood	qPCR	sNANA 5s	1	Down	miR-219a-5p	NA	IBD type, disease severity
Swati Valmiki 2020	UC	66.67/33.33	37.02±11.87	48	Non-IBD	80/20	38.25±13.91	30	Mucosa	qPCR	U6	2	Up	miR-125b, miR-223	miR-125bT RAF6 and A20, miR-223: IKKα	NA
Dandan Zhao 2020	IBD	NA	NA	30	Non-IBD	NA	NA	30	Mucosa	qPCR	U6	1	Up	miR-449a	Notch 1	NA
FriederikeCordes 2020	IBD	NA	NA	UC: 37, CD: 40	Healthy controls	NA	NA	19	Blood	qPCR	NA	1	Up	miR-320a	NA	Disease severity and treatment response
Xiaozhi Deng 2020	IBD	50/50	9.4 (7-11)	30	Non-IBD	50/50	8.8 (6-12)	30	Mucosa	qPCR	U6	1	Up	miR-4262	SIRT1	NA
Richard K. Felwick 2020	CD	47.83/52.17	38 (17-80)	23	Healthy controls	30/70	70 (49-86)	10	Mucosa	qPCR	RNU44	1	Up	miR-23a	TNFAIP3	NA
Małgorzata Guz 2020	IBD	UC: 62.5/38.5, CD:50/50	UC: 49.2 (23-27), CD:45.3 (23-67)	UC: 16, CD: 12	Healthy controls	45.45/54.55	59.4 (22-88)	11	Mucosa	qPCR	NA	5	Up	miR-21-3p, miR-31-3p, miR-125b-1-3p, miR-146a-3p, miR-155-5p	NA	NA
Agnieszka Kempinska-Podhorodecka 2020	UC	20/80	43.5±15	10	Healthy controls	60/40	50±4	10	Mucosa	qPCR	NA	1	Down	miR-346	NA	Primary sclerosing cholangitis
Mohammad Mirzakhani 2020	CD	52.17/47.83	37.04±10.45	23	Healthy controls	50/50	36.9±10.6	30	Blood	qPCR	RNU48	2	Up	miR-21, miR-146a	NA	Disease severity

Mehri Naghdalipour 2020	UC	52.2/47.8	58.09±6.27	23	Non-IBD	38.9/61.1	56.11±9.93	18	Mucosa	qPCR	miR-361-5p	3	Up or down	Up: miR-21, miR-433, down: miR-590	NA	NA
Xiaojuan Shao 2020	IBD	62/38	NA	50	Healthy controls	50/50	NA	24	Blood	qPCR	NA	1	Up	miR-155	GPER1	Gender
Jun Yao 2020	IBD	UC: 42/58, CD: 54/46	UC: 35.1±6.45, CD: 32.8±7.3	UC: 50, CD:50	Healthy controls	55/45	35±8.1	20	Mucosa and PBMC	qPCR	U6	1	Up	miR-802	SOCS5	Disease severity
Ting Yu 2020	Active UC	61.29/38.71	38.6±6.2	62	Healthy controls	61.29/38.71	38.7±6.1	62	Plasma	qPCR	NA	1	Up	miR-24	NA	NA
Xiaojing Zhao 2020	CD	NA	NA	10	Healthy controls	NA	NA	8	Mucosa	qPCR	U6	1	Up	miR-124a	AHR	NA

miRNA: microRNAs; IBD: Inflammatory bowel disease; UC: Ulcerative colitis; CD: Crohn's disease; y: years; SD: Standard deviation; PBMC: Peripheral blood mononuclear cell; FFPE: Formalin-fixed paraffin-embedded; qPCR: Quantitative real-time polymerase chain reaction; NA: Not available; Non: None of valid data.

Table S4 Consistently up-regulated and down-regulated miRNAs that were reported in studies with microarray analysis.

	miRNA	No. of studies with same direction	Total number of IBD samples tested
Up-regulated miRNA	miR-223	10	376
	miR-21	7	60
	miR-155	3	35
	miR-708-5p	3	33
	miR-142	2	66
	miR-1246	2	64
	miR-16	2	60
	miR-27a	3	45
	miR-15b	2	39
	miR-144	2	25
	miR-30b	2	19
	miR-1973	2	15
	miR-451	2	13
	miR-106a	2	12
	miR-17	2	10
	miR-126	2	9
	miR-106b	2	11
	miR-15a	2	8
Down-regulated miRNA	miR-192-5p	3	64
	miR-4286	2	11
	miR-4284	2	11

miRNA: microRNAs; No.: Number.

Table S5 Consistently up-regulated and down-regulated miRNAs that were reported in studies with validated methods.

	miRNA	No. of studies with same direction	Total number of IBD samples tested
Up-regulated miRNA	miR-155	11	493
	miR-31	9	429
	miR-21	9	396
	miR-223	5	121
	miR-126	4	276
	miR-23a	3	262
	miR-122	3	70
	miR-146a	3	49
	miR-301a	2	291
	miR-362	2	280
	miR-106a	2	280
	miR-143	2	234
	miR-16	2	226
	miR-24	2	141
	miR-320a	2	92
	miR-146	2	60
	miR-124	2	30
	miR-132	2	43
	miR-206	2	41
Down-regulated miRNA	miR-141	4	92
	miR-200b	3	84
	miR-200a	2	62
	miR-200c	2	62

miRNA: microRNAs; No.: Number.

Table S6 Results of meta-regression analysis.

Var	Coeff.	Std. Err.	P value	RDOR	95%CI
Cte.	-0.741	1.0584	0.4895	—	—
S	-0.15	0.145	0.3103	—	—
IBD type	-0.639	0.3007	0.0419	0.53	(0.29 - 0.98)
Age	0.558	0.4115	0.1856	1.75	(0.75 - 4.05)
Sample	1.473	0.6325	0.0268	4.36	(1.20 - 15.88)
miRNAassay	1.861	0.5804	0.0032	6.43	(1.96 - 21.03)
Control	1.587	0.3876	0.0003	4.89	(2.22 - 10.79)

No. studies =37

Filter OFF

Add 1/2 to all cells of the studies with zero

IBD type: UC versus CD, age of participants: pediatric population (<18 years of age) versus adult (⩾18 years of age), sample source: blood versus others, method of quantifying miRNA expression (miRNAassay): qPCR versus only microarray and control: healthy controls versus non-IBD controls.

Abbreviations: miRNA = microRNAs; CI = confidence interval; Coeff. = coefficient; Std. Err. = standard error; RDOR = relative diagnostic odds ratios.

Tau-squared estimate = 0.5749 (Convergence is achieved after 9 iterations)

Restricted Maximum Likelihood estimation (REML)

Table S7 Results of sensitivity analysis.

StudyID	Sensitivity (95%CI)	Specificity (95%CI)	PLR (95%CI)	NLR (95%CI)	DOR (95%CI)
Adam M. Zahm 2011	0.82 (0.80-0.84)	0.83 (0.80-0.85)	4.03 (3.03-5.37)	0.24 (0.17-0.34)	21.06 (11.80-37.57)
Adam M. Zahm 2014	0.81 (0.79-0.82)	0.84 (0.82-0.86)	4.71 (3.64-6.10)	0.26 (0.21-0.33)	23.45 (14.93-36.84)
Chenming Sun 2017	0.81 (0.79-0.82)	0.85 (0.83-0.87)	4.44 (3.55-5.54)	0.26 (0.21-0.32)	22.43 (14.47-34.77)
Christos Polytarchou 2015	0.81 (0.79-0.83)	0.85 (0.83-0.87)	4.96 (3.80-6.46)	0.25 (0.20-0.32)	25.50 (16.21-40.12)
Katharina Schönauen 2018	0.80 (0.79-0.82)	0.84 (0.82-0.86)	4.41 (3.46-5.63)	0.26 (0.21-0.33)	21.20 (13.77-32.62)
Matthias Hübenthal 2015	0.80 (0.78-0.81)	0.83 (0.81-0.85)	4.20 (3.33-5.30)	0.27 (0.22-0.33)	19.50 (13.02-29.19)
Mengdie Shen 2019	0.80 (0.78-0.82)	0.84 (0.82-0.86)	4.51 (3.51-5.80)	0.27 (0.22-0.33)	21.37 (13.80-33.08)
Michael D. Jensen 2015	0.81 (0.80-0.83)	0.84 (0.82-0.86)	4.58 (3.59-5.84)	0.26 (0.21-0.32)	22.43 (14.88-33.81)
Olfat G. Shaker 2019	0.80 (0.79-0.82)	0.83 (0.81-0.85)	4.28 (3.40-5.40)	0.27 (0.22-0.33)	19.88 (13.16-30.04)
Peng Chen 2019 (training cohort）	0.81 (0.79-0.83)	0.84 (0.82-0.86)	4.49 (3.49-5.77)	0.26 (0.21-0.33)	22.10 (14.07-34.73)
Peng Chen 2019 (validation cohort)	0.81 (0.80-0.83)	0.84 (0.82-0.86)	4.38 (3.43-5.58)	0.26 (0.21-0.32)	21.54 (13.95-33.27)
Radha Duttagupta 2012	0.80 (0.78-0.82)	0.84 (0.82-0.86)	4.36 (3.42-5.56)	0.27 (0.22-0.33)	20.40 (13.37-31.12)
Yulan Ye 2017	0.80 (0.79-0.82)	0.84 (0.82-0.86)	4.54 (3.55-5.81)	0.26 (0.21-0.33)	21.96 (14.24-33.86)
Elham Ahmed Hassan 2020	0.80 (0.78-0.82)	0.83 (0.81-0.85)	4.18 (3.31-5.28)	0.27 (0.22-0.34)	19.23 (12.65-29.24)
Yuanyuan Tian 2020	0.76 (0.74-0.78)	0.85 (0.83-0.87)	4.52 (3.44-5.94)	0.30 (0.26-0.35)	17.08 (11.81-24.71)
Rui Zhou 2021	0.81 (0.79-0.82)	0.84 (0.82-0.86)	4.36 (3.39-5.62)	0.26 (0.21-0.33)	21.46 (13.64-33.76)

95%CI: 95% confidence interval; PLR: Positive likelihood ratio; NLR: Negative likelihood ratio;DOR: Diagnostic odds ratio.